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Three dimensional molding device

Abstract

This three dimensional molding device includes a plasticizing section having a screw in which a groove is formed and generating a plasticized material by plasticizing a material by rotating the screw; a nozzle having a nozzle opening and ejecting the plasticized material; an ejection adjustment section provided in a flow path, which communicates with the nozzle opening, and adjusting an ejecting amount of the plasticized material from the nozzle by adjusting an opening area of the flow path; and a control section, wherein the control section controls a rotation speed of the screw at a first rotation speed when the opening area is a first opening area and controls the rotation speed of the screw at a second rotation speed that is faster than the first rotation speed when the opening area is a second opening area that is larger than the first opening area.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-157651, filed Sep. 30, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a three dimensional molding device and a method for manufacturing a three dimensional molded object.

2. Related Art

(3) JP-A-2022-16532 discloses a three dimensional molding device including a butterfly valve as a flow amount adjustment mechanism for adjusting the ejecting amount of a molten material from a nozzle.

(4) When the ejecting amount is decreased by a flow amount adjustment mechanism such as a butterfly valve, pressure in the flow path upstream of the flow amount adjustment mechanism increases as time elapses. Therefore, when the ejecting amount is increased by the flow amount adjustment mechanism after the ejecting amount is decreased, an unexpected ejecting amount of

molten material is ejected from the nozzle due to increased pressure, molding accuracy is affected.

SUMMARY

(5) According to a first aspect of the present disclosure, a three dimensional molding device is provided.

(6) This three dimensional molding device includes a plasticizing section that has a screw in which a groove is formed and that is configured to generate a plasticized material by plasticizing a material by rotating the screw; a nozzle that has a nozzle opening and that is configured to eject the plasticized material; a table on which the plasticized material ejected from the nozzle is deposited; a position changing section configured to change a relative position between the nozzle and the table; an ejection adjustment section that is provided in a flow path, which communicates with the nozzle opening and through which the plasticized material flows, and that is configured to adjust an ejecting amount of the plasticized material from the nozzle by adjusting an opening area of the flow path; and a control section configured to mold a three dimensional molded object in a molding area of the table by controlling the plasticizing section, the position changing section, and the ejection adjustment section, wherein the control section controls a rotation speed of the screw at a first rotation speed when the opening area of the flow path is a first opening area and controls the rotation speed of the screw at a second rotation speed that is faster than the first rotation speed when the opening area of the flow path is a second opening area that is larger than the first opening area.

(7) According to a second aspect of the present disclosure, there is provided a method for manufacturing a three dimensional molded object.

(8) This manufacturing method includes a first step of plasticizing a material by rotating a screw, in which a groove is formed, to generate a plasticized material and a second step of molding a three dimensional molded object by discharging the plasticized material from a nozzle having a nozzle opening, wherein the second step includes a step of adjusting an ejecting amount of the plasticized material from the nozzle by adjusting an opening area of a flow path which communicates with the nozzle opening and through which the plasticized material flows and the first step includes a step of setting the rotation speed of the screw to a first rotation speed when the opening area of the flow path is a first opening area and setting the rotation speed of the screw to a second rotation speed that is higher than the first rotation speed when the opening area of the flow path is second opening area that is larger than the first opening area.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is an explanatory view showing a schematic configuration of a three dimensional molding device according to a first embodiment.

(2) FIG. 2 is a perspective view showing a schematic configuration of a flat screw.

(3) FIG. 3 is a schematic plan view of a barrel.

(4) FIG. 4 is a perspective view showing a schematic configuration of a butterfly valve.

(5) FIG. 5 is an explanatory view schematically showing a basic operation in which the three dimensional molding device molds a molded object.

(6) FIG. 6 is a flowchart of a three dimensional molding process in the first embodiment.

(7) FIG. 7 is a graph showing the relationship between an opening area and a rotation speed of the flat screw.

(8) FIG. 8 is a timing chart showing control timings of the flat screw and the butterfly valve.

(9) FIG. 9 is a flowchart of the three dimensional molding process according to a second embodiment.

DESCRIPTION OF EMBODIMENTS

A. First Embodiment

(10) FIG. 1 is an explanatory view showing a schematic configuration of a three dimensional molding device **100** according to a first embodiment. In FIG. 1, arrows indicating X, Y, and Z directions orthogonal to each other are shown. The X direction and the Y direction are directions parallel to a horizontal plane, and the Z direction is a direction along a vertically upward direction. The arrows indicating the X, Y, and Z directions are appropriately illustrated in other drawings so that the illustrated directions correspond to those in FIG. 1. In the following description, when a direction to be specified, a direction indicated by an arrow in each drawing is referred to as “+” and an opposite direction is referred to as “-”, and positive and negative signs are used in combination for direction notation. Hereinafter, the +Z direction is also referred to as “upper”, and the -Z direction is also referred to as “lower”.

(11) The three dimensional molding device **100** of the present embodiment is a device that molds a molded object by a material extrusion method. The three dimensional molding device **100** includes a molding section **110** that generates and ejects a plasticized material, a table **210** for molding serving as a base of a molded object, a position changing section **230** that controls an eject position of the plasticized material, and a control section **300** that controls each unit of the three dimensional molding device **100**.

(12) Under the control of the control section **300**, the molding section **110** ejects a plasticized material obtained by plasticizing a material in a solid state onto the table **210**. The molding section **110** includes a material supply section **20** that is a supply source of a raw material before being converted into the plasticized material, a plasticizing section **30** that converts the raw material into the plasticized material, and an ejection section **60** that ejects the plasticized material.

(13) The material supply section **20** supplies a raw material MR to the plasticizing section **30**. The material supply section **20** is constituted by, for example, a hopper for containing the raw material MR. The material supply section **20** is connected to the plasticizing section **30** via a communication path **22**. The raw material MR is supplied to the material supply section **20** in the form of pellets, powder, or the like. As the raw material, for example, a resin material such as ABS (acrylonitrile-butadiene-styrene), PEEK (polyetheretherketone), or PP (polypropylene) is used.

(14) The plasticizing section **30** plasticizes the raw material MR supplied from the material supply section **20** to generate a pasty plasticized material exhibiting fluidity, and guides the plasticized material to the ejection section **60**. In the present embodiment, “plasticization” is a concept that includes melting and is a change from a solid to a fluid state. Specifically, in the case of a material that undergoes a glass transition, plasticization means that the temperature of the material is set to be equal to or higher than its glass transition point. In the case of a material that does not undergo glass transition, plasticization means that the temperature of the material is set to be equal to or higher than its melting point.

(15) The plasticizing section **30** includes a screw case **31**, a drive motor **32**, a flat screw **40**, and a barrel **50**. The flat screw **40** is also referred to as a rotor or simply a screw. The barrel **50** is also referred to as a screw facing section.

(16) FIG. 2 is a perspective view showing a schematic configuration of a lower surface **48** side of the flat screw **40**. In order to facilitate understanding of the technology, the flat screw **40** shown in FIG. 2 is shown in a state in which the positional relationship between an upper surface **47** and the lower surface **48** shown in FIG. 2 is reversed in the vertical direction. FIG. 3 is a schematic plan view showing the upper surface **52** side of the barrel **50**. The flat screw **40** has a substantially columnar shape in which a length in an axial direction, which is a direction along a central axis, is smaller than a length in a direction perpendicular to the axial direction. The flat screw **40** is disposed such that a rotation axis RX, which is its center of rotation, is parallel to the Z direction.

(17) As shown in FIG. 1, the flat screw **40** is housed in the screw case **31**. The upper surface **47** of the flat screw **40** is connected to the drive motor **32**, and the flat screw **40** is rotated in the screw case **31** by a rotational driving force generated by the drive motor **32**. The rotation speed of the flat

screw **40** is controlled by the control section **300**. The flat screw **40** may be driven by the drive motor **32** via a reduction gear.

(18) As shown in FIG. 2, vortex shaped groove sections **42** are formed on the lower surface **48** of the flat screw **40**, which is a surface intersecting the rotation axis RX. The communication path **22** of the material supply section **20** described above communicates with the groove sections **42** from the side surface of the flat screw **40**. In the present embodiment, the groove sections **42** are separated from each other by the ridge sections **43**, and three groove sections **42** are formed. The number of the groove sections **42** is not limited to three, and may be one or two or more. The groove sections **42** are not limited to a vortex shape, but may be a spiral shape or an involute curve shape, or may be a shape extending so as to draw an arc from the central section toward the outer periphery.

(19) The lower surface **48** of the flat screw **40** faces the upper surface **52** of the barrel **50**, and space is formed between the groove sections **42** of the lower surface **48** of the flat screw **40** and the upper surface **52** of the barrel **50**. The raw material MR is supplied to the space between the flat screw **40** and the barrel **50** from the material supply section **20** through a material inflow port **44** (shown in FIG. 2).

(20) As shown in FIG. 1, a barrel heater **58** for heating the raw material MR supplied into the groove sections **42** of the rotating flat screw **40** is embedded in the barrel **50**. A communication hole **56** is provided at the center of the barrel **50**. As shown in FIG. 3, the upper surface **52** of the barrel **50** is formed with a plurality of guide grooves **54** that are connected to the communication hole **56** and that extend vortically from the communication hole **56** toward the outer periphery. One end of the guide grooves **54** may not be connected to the communication hole **56**. The guide grooves **54** may be omitted.

(21) The raw material MR supplied into the groove sections **42** of the flat screw **40** flows along the groove sections **42** by the rotation of the flat screw **40** while being plasticized in the groove sections **42**, and is guided to a central section **46** of the flat screw **40** as a plasticized material. The plasticized material in a paste state exhibiting fluidity and which flowed into the central section **46** is supplied to the ejection section **60** via the communication hole **56** provided at the center of the barrel **50**. In the plasticized material, not all types of substances constituting the plasticized material need be plasticized. The plasticized material may be converted into a state having fluidity as a whole by plasticizing at least some types of substances among the substances constituting the plasticized material.

(22) The ejection section **60** of FIG. 1 includes a nozzle **61** that ejects the plasticized material from a nozzle opening **62**, a flow path **65** of the plasticized material provided between the flat screw **40** and the nozzle opening **62**, and an ejection control section **77** that controls ejection of the plasticized material.

(23) The nozzle **61** is connected to the communication hole **56** of the barrel **50** through the flow path **65**. The nozzle **61** ejects the plasticized material generated in the plasticizing section **30** toward the table **210** from the nozzle opening **62** at the tip end.

(24) The ejection control section **77** includes an ejection adjustment section **70** that opens and closes the flow path **65** and a suction section **75** that sucks and temporarily stores the plasticized material.

(25) The ejection adjustment section **70** is provided in the flow path **65** which communicates with the nozzle opening **62** and through which the plasticized material flows. The ejection adjustment section **70** adjusts the ejecting amount of the plasticized material from the nozzle **61** by adjusting the opening area of the flow path **65**. The communication hole **56** formed in the barrel **50** is a part of the flow path **65**.

(26) The ejection adjustment section **70** includes a drive shaft **71**, a butterfly valve **72**, an angle detection sensor **73**, and a first drive section **74**.

(27) FIG. 4 is a perspective view showing a schematic configuration of the butterfly valve **72**. The

drive shaft **71** is a shaft disposed along a direction intersecting the direction in which the flow path **65** extends. In the present embodiment, the flow path **65** extends along the Z direction, and the drive shaft **71** is disposed along the Y direction. The butterfly valve **72** is formed on a part of the drive shaft **71** in the flow path **65**. The butterfly valve **72** is a member obtained by processing a part of the drive shaft **71**. The butterfly valve **72** is rotatably disposed in the flow path **65**. The drive shaft **71** is provided so that the butterfly valve **72** is located at a position where the drive shaft **71** and the flow path **65** intersect with each other. The rotation angle of the butterfly valve **72** shown in FIG. **4** is an angle at which the opening area of the flow path **65** is zero. The shape of the butterfly valve **72** is not limited as long as it rotates in the flow path **65** to adjust the opening area of the flow path **65**, and may be, for example, a plate shape or a hemispherical shape. The butterfly valve **72** is also referred to as a valve.

(28) The angle detection sensor **73** shown in FIG. **1** is a sensor for detecting the rotation angle of the butterfly valve **72**. As the angle detection sensor **73**, for example, a rotary encoder or a resolver can be used.

(29) The first drive section **74** rotates the butterfly valve **72** by rotating the drive shaft **71**. The first drive section **74** is configured by, for example, a stepping motor. The control section **300** adjusts the opening area of the flow path **65** by controlling the rotation angle of the butterfly valve **72** using the first drive section **74**, and adjusts the flow amount of the plasticized material flowing from the plasticizing section **30** to the nozzle **61**, that is, the ejecting amount of the plasticized material ejected from the nozzle **61**. The control section **300** executes feedback control of the rotation angle of the butterfly valve **72** using the detection value of the angle detection sensor **73**. The ejection adjustment section **70** can adjust the ejecting amount of the plasticized material and can control ON/OFF of the outflow of the plasticized material.

(30) In the adjustment of the opening area of the flow path **65** using the ejection adjustment section **70**, the control section **300** can execute either a process of changing the opening area at a first adjustment speed or a process of changing the opening area at a second adjustment speed that is higher than the first adjustment speed. That is, the control section **300** can adjust the opening area of the flow path **65** at least two stages of adjustment speeds. In the present embodiment, the process of changing the opening area is a process of driving the drive shaft **71** to rotate the butterfly valve **72**, and the adjustment speed is the rotation speed of the butterfly valve **72**.

(31) The suction section **75** is connected between the ejection adjustment section **70** and the nozzle opening **62** in the flow path **65**. The suction section **75** temporarily sucks the plasticized material in the flow path **65** when ejection of the plasticized material from the nozzle **61** is stopped, thereby suppressing a tailing phenomenon in which the plasticized material drips threadlike from the nozzle opening **62**. In the present embodiment, the suction section **75** is constituted by a plunger. The suction section **75** is driven by the second drive section **76** under the control of the control section **300**. The second drive section **76** is configured by, for example, a stepping motor, a rack and pinion mechanism that converts rotational force of the stepping motor into translational motion of a plunger, or the like.

(32) A pressure sensor **80** is provided in the flow path **65**. The pressure sensor **80** is provided in the flow path **65** upstream of the ejection adjustment section **70**. The pressure sensor **80** detects the pressure in the flow path **65** upstream of the ejection adjustment section **70**.

(33) The table **210** is disposed at a position facing the nozzle opening **62** of the nozzle **61**. In the first embodiment, the molding surface **211** of the table **210** facing the nozzle opening **62** of the nozzle **61** is disposed to be parallel to the X and Y directions, that is, a horizontal direction. The table **210** may be provided with a heater for suppressing rapid cooling of the plasticized material ejected onto the table **210**.

(34) The position changing section **230** changes the relative position between the table **210** and the nozzle **61** under the control of the control section **300**. In the present embodiment, the position of the nozzle **61** is fixed, and the position changing section **230** moves the table **210**. The position

changing section **230** is configured by a three axis positioner that moves the table **210** in three axial directions of X, Y, and Z directions by driving forces of three motors. In the present specification, unless otherwise specified, the movement of the nozzle **61** means that the nozzle **61** or the ejection section **60** is moved relative to the table **210**.

(35) In another embodiment, instead of the configuration in which the table **210** is moved by the position changing section **230**, a configuration may be employed in which the position changing section **230** moves the nozzle **61** with respect to the table **210** in a state in which the position of the table **210** is fixed. In addition, a configuration may be employed in which the table **210** is moved in the Z direction by the position changing section **230** and the nozzle **61** is moved in the X and Y directions, or a configuration in which the table **210** is moved in the X and Y directions by the position changing section **230** and the nozzle **61** is moved in the Z direction. Even in these configurations, the relative positional relationship between the nozzle **61** and the table **210** can be changed.

(36) The control section **300** is configured by a computer that has one or a plurality of processors **310**, a storage section **320** that includes a main storage section and an auxiliary storage device, and an input/output interface that inputs and outputs signals to and from the outside. By executing a program stored in the storage section **320**, the processor **310** controls the plasticizing section **30**, the position changing section **230**, and the ejection control section **77** according to molding data stored in the storage section **320**, thereby molding the three dimensional molded object in the molding region of the table **210**. The molding data for molding the three dimensional molded object includes, for each layer of the molded object shaped sliced into a plurality of layers, path information representing a movement path of the nozzle **61** and ejecting amount information indicating the ejecting amount of the plasticized material in each movement path. The movement path of the nozzle **61** is a path in which the nozzle **61** relatively moves along the molding surface **211** of the table **210** while ejecting the plasticized material. The control section **300** may be realized by a configuration in which circuits are combined, instead of being configured by a computer.

(37) FIG. 5 is an explanatory view schematically showing a basic operation in which the three dimensional molding device **100** molds the molded object. In the three dimensional molding device **100**, as described above, the raw material MR in a solid state is plasticized to generate the plasticized material MM. The control section **300** causes the nozzle **61** to eject plasticized material MM while changing the position of the nozzle **61** with respect to the table **210** in a direction along the molding surface **211** of the table **210** while maintaining the distance between the molding surface **211** of the table **210** and the nozzle **61**. The plasticized material MM ejected from the nozzle **61** is continuously deposited in the moving direction of the nozzle **61**.

(38) The control section **300** repeats the movement of the nozzle **61** to form a molded layer ML. After forming one molded layer ML, the control section **300** relatively moves the position of the nozzle **61** with respect to the table **210** in the Z direction. Then, the molded object is formed by stacking further molded layers ML on the molded layers ML formed so far.

(39) For example, in a case where the nozzle **61** is to be moved in the Z direction when one layer worth of the molded layers ML is completed, or in a case where there are a plurality of independent molding regions in each molded layer, in some case, the control section **300** temporarily stops ejection of the plasticized material from the nozzle **61**. In this case, the flow path **65** is closed by the ejection adjustment section **70** to stop ejection of the plasticized material MM from the nozzle opening **62**, and the plasticized material in the nozzle **61** is temporarily sucked by the suction section **75**. After changing the position of the nozzle **61**, the control section **300** causes the ejection adjustment section **70** to open the flow path **65** while ejecting the plasticized material in the suction section **75**, thereby resuming the deposition of the plasticized material MM from the changed position of the nozzle **61**.

(40) FIG. 6 is a flowchart of a three dimensional molding process executed by the control section **300**. A method of manufacturing the three dimensional molded object is realized by executing the

three dimensional molding process. While the three dimensional molding process is being executed, the control section **300** controls the plasticizing section **30**, the position changing section **230**, and the ejection adjustment section **70** in accordance with the molding data, thereby depositing the plasticized material on the table **210** and molding the three dimensional molded object.

(41) In step **S10**, the control section **300** determines the opening area of the flow path **65** based on the ejecting amount information in the molding data. Specifically, the larger the ejecting amount, the larger the value that the control section **300** sets the opening area of the flow path **65** and the smaller the ejecting amount, the smaller the value that the control section **300** sets the opening area of the flow path **65**.

(42) In step **S12**, the control section **300** determines the rotation speed of the flat screw **40** based on the opening area determined in step **S10**.

(43) FIG. 7 is a graph showing the relationship between the opening area of the flow path **65** and the rotation speed of the flat screw **40**. As shown in FIG. 7, the larger the opening area of the flow path **65**, the more the control section **300** increases the rotation speed of the flat screw **40**. For example, when the opening area of the flow path **65** is a second opening area **S2**, which is larger than a first opening area **S1**, the control section **300** determines the rotation speed of the screw case **31** is to be a second rotation speed **R2**, which is faster than a first rotation speed **R1**. The relationship between the opening area of the flow path **65** and the rotation speed of the flat screw **40** is determined in advance through experiments and simulations so that the amount of the plasticized material supplied to the ejection adjustment section **70** and the amount of the plasticized material flowing out from the ejection adjustment section **70** do not become excessive or deficient when the opening area of the flow path **65** is changed by the ejection adjustment section **70**.

(44) In step **S14** of FIG. 6, the control section **300** controls the plasticizing section **30** to adjust the rotation speed of the flat screw **40** to the rotation speed determined in step **S12**.

(45) After the adjustment of the rotation speed of the flat screw **40**, then in step **S16**, the control section **300** controls the ejection adjustment section **70** to adjust the opening area of the flow path **65** to the opening area determined in step **S10**.

(46) FIG. 8 is a timing chart showing control timings of the flat screw **40** and the butterfly valve **72**. According to the above described step **S14** and step **S16**, in the present embodiment the control section **300** starts an operation for changing the rotation speed of the flat screw **40** earlier than an operation for changing the opening area of the flow path **65**. In the timing chart shown in FIG. 8, the control section **300** starts the operation for changing the rotation speed of the flat screw **40** at timing **T1**, and starts the operation for changing the rotation angle of the butterfly valve **72** at timing **T2**, which is later than timing **T1**. The difference between the timing **T1** at which the rotation speed of the flat screw **40** is changed and the timing **T2** at which the opening area of the flow path **65** is subsequently changed is desirably set by executing experiments or simulations in advance to be a timing at which the plasticized material, in a generation amount that was increased or decreased by changing the rotation speed of the flat screw **40**, passes through the flow path **65** and reaches the butterfly valve **72** when the rotation angle of the butterfly valve **72** is changed.

(47) It is desirable that the change of the rotation speed of the flat screw **40** in step **S14** and the change of the opening area of the flow path **65** in step **S16** are executed so that the times required for the changes coincide with each other. In FIG. 8, the time **Tm1** required for changing the rotation speed of the flat screw **40** and the time **Tm2** required for changing the angle of the butterfly valve **72** coincide with each other. Thus, when the angle of the butterfly valve **72** is changed, excess or deficiency of the plasticized material supplied to the butterfly valve **72** can be suppressed.

(48) In step **S18** of FIG. 6, the control section **300** judges whether the pressure in the flow path **65** measured by the pressure sensor **80** exceeds a predetermined threshold. When it is determined that the pressure exceeds the predetermined threshold value, then in step **S20**, the control section **300** decreases the rotation speed of the flat screw **40**. Specifically, the control section **300** stops the

rotation of the flat screw **40** until the pressure measured by the pressure sensor **80** falls below the threshold described above. In this way, “reducing the rotation speed” includes stopping the rotation. When it is judged in step **S18** that the pressure does not exceed the predetermined threshold, the control section **300** skips the process of step **S20**. Note that the control section **300** may make the judgement in step **S18** by using a mean value of one or more pressure measurement values measured in the past and the pressure measurement value measured this time. In this way, even in a case where a measurement error is included in the measurement value due to the influence of noise or the like, it is possible to appropriately determine whether or not to decrease the rotation speed of the flat screw **40**.

(49) In step **S22**, the control section **300** judges whether or not the movement speed of the nozzle **61** should be decreased. For example, when the nozzle **61** is moved along a bent path, the control section **300** decreases the movement speed of the nozzle **61**. When it is judged that the movement speed of the nozzle **61** is to be decreased, then in step **S24**, the control section **300** sets the rotation speed of the butterfly valve **72** to the first adjustment speed and rotates the butterfly valve **72** at the first adjustment speed, thereby reducing the opening area of the flow path **65** from the second opening area to the first opening area and reducing the ejecting amount of the plasticized material. The process of step **S24** is also referred to as a first process. In step **S22**, when it is not judged that the movement speed of the nozzle **61** is to be decreased, the control section **300** skips the process of step **S24**.

(50) In step **S26**, the control section **300** judges whether to increase the movement speed of the nozzle **61**. For example, the control section **300** increases the movement speed of the nozzle **61** when the nozzle **61** is moved along a linear path. When it is judged that the movement speed of the nozzle **61** is to be increased, then in step **S28**, the control section **300** sets the rotation speed of the butterfly valve **72** to the second adjustment speed, which is higher than the first adjustment speed, and rotates the butterfly valve **72** at the second adjustment speed, thereby increasing the opening area of the flow path **65** from the first opening area to the second opening area and increasing the ejecting amount of the plasticized material. The process of step **S28** is also referred to as a second process. In step **S26**, when it is not judged to increase the movement speed of the nozzle **61**, then the control section **300** skips the process of step **S28**. The judgement of whether or not to change the movement speed of the nozzle **61** in step **S22** and step **S26** described above may be executed based on, for example, a command or data for changing the movement speed included in the molding data.

(51) In step **S30**, the control section **300** judges whether or not the molding of the three dimensional molded object is completed. When it is judged that the molding of the three dimensional molded object is completed, then the control section **300** terminates the three dimensional molding process. On the other hand, in a case where it is judged that the molding of the three dimensional molded object is not completed, then the control section **300** returns the process to step **S10** and continues the molding of the three dimensional molded object.

(52) In the three dimensional molding device **100** according to the first embodiment described above, as shown in FIG. 7, the rotation speed of the flat screw **40** is controlled at the first rotation speed when the opening area of the flow path **65** is the first opening area, and the rotation speed of the flat screw **40** is controlled at the second rotation speed, which is higher than the first rotation speed, when the opening area of the flow path **65** is the second opening area, which is larger than the first opening area. Therefore, when the ejecting amount of the plasticized material from the nozzle **61** is large, the rotation speed of the flat screw **40** can be increased to increase the generation amount of the plasticized material, and when the ejecting amount of the plasticized material from the nozzle **61** is small, the rotation speed of the flat screw **40** can be decreased to decrease the generation amount of the plasticized material. Therefore, for example, in the three dimensional molding device in which the rotation speed of the flat screw **40** is set to a constant speed, when the opening area of the flow path **65** is decreased, the pressure in the flow path **65**

upstream of the ejection adjustment section **70** increases, and when the opening area of the flow path **65** is subsequently increased, there is a possibility that a phenomenon occurs in which an unexpected ejecting amount of plasticized material is ejected from the nozzle **61**, but such a phenomenon can be suppressed in the present embodiment. Therefore, the molded object can be molded with high accuracy.

(53) In addition, in the present embodiment, since the ejection adjustment section **70** which adjusts the ejecting amount of the plasticized material from the nozzle **61** includes the butterfly valve **72**, and the control section **300** controls the rotation angle of the butterfly valve **72** based on the detection value of the angle detection sensor **73**, it is possible to adjust the opening area of the flow path **65** with a simple configuration.

(54) Further, in the present embodiment, when the pressure, which was measured by the pressure sensor **80** detecting the pressure in the flow path **65** upstream of the butterfly valve **72**, exceeds a predetermined value, then the rotation speed of the flat screw **40** is decreased. Therefore, it is possible to prevent the pressure in the flow path **65** upstream of the butterfly valve **72** from increasing excessively, and it is possible to more reliably prevent the plasticized material of an unintended ejecting amount from being ejected from the nozzle **61**. In addition, by decreasing the rotation speed of the flat screw **40**, it is possible to suppress the resin from being modified in the plasticizing section **30**.

(55) Further, in the present embodiment, the control section **300** starts the operation for changing the rotation speed of the flat screw **40** earlier than the operation for changing the opening area of the flow path **65**. Therefore, when the opening area of the flow path **65** is changed, it is possible to suppress the occurrence of insufficient supply of the plasticized material from the plasticizing section **30** to the ejection adjustment section **70**, and it is possible to suppress the occurrence of ejection defects.

(56) In addition, in the present embodiment, in the adjustment of the opening area of the flow path **65** using the ejection adjustment section **70**, the control section **300** executes either a process of changing the opening area at a first adjustment speed and a process of changing the opening area at a second adjustment speed faster than the first adjustment speed. Therefore, the control section **300** can adjust the opening area of the flow path **65** in at least two stages of adjustment speeds.

(57) In addition, in the present embodiment, in the adjustment of the opening area, the control section **300** can execute either the first process of changing from the second opening area to the first opening area at the first adjustment speed or the second process of changing from the first opening area to the second opening area at the second adjustment speed. Then, the control section **300** executes the first process in a case where the relative movement speed of the nozzle **61** is to be decreased and executes the second process in a case where the relative movement speed of the nozzle **61** is to be increased. In other words, according to the present embodiment, the opening area of the flow path **65** is decreased when the relative movement speed of the nozzle **61** is to be decreased and the opening area of the flow path **65** is increased when the relative movement speed of the nozzle **61** is to be increased. The adjustment speed of the opening area is made faster when the opening area of the flow path **65** is to be increased than when it is to be decreased. Therefore, when the movement speed of the nozzle **61** is to be increased, it is possible to rapidly increase the ejecting amount of the plasticized material, and it is possible to suppress a line width of the plasticized material from becoming thin in the linear section of the molded object.

B. Second Embodiment

(58) FIG. **9** is a flowchart of the three dimensional molding process according to a second embodiment. In the second embodiment, the process in step **S18b** shown in FIG. **9** is different from the process in step **S18** in the first embodiment, and the other processing contents are the same as those in the first embodiment. The configuration of the three dimensional molding device **100** in the second embodiment is the same as that in the first embodiment. However, in the second embodiment, the three dimensional molding device **100** may not include the pressure sensor **80**.

(59) As shown in FIG. 9, in the second embodiment, after the process from step S10 to step S16 is finished, in step S18b, the control section 300 judges whether or not a state in which the opening area of the flow path 65 is equal to or less than a predetermined value has continued for a predetermined time or more. When it is judged that the state in which the opening area of the flow path 65 is equal to or less than the predetermined value has continued for the predetermined time or more, then in step S20 the control section 300 decreases the rotation speed of the flat screw 40 to decrease the pressure in the flow path 65 upstream of the butterfly valve 72.

(60) According to the second embodiment described above, it is possible to suppress the pressure in the flow path 65 upstream of the butterfly valve 72 from excessively increasing without using the pressure sensor 80. Therefore, similarly to the first embodiment, it is possible to more reliably suppress the plasticized material of an unintended ejecting amount from being ejected from the nozzle 61. By executing experiments or simulations, the above described “predetermined value” and “predetermined time” can be set according to whether or not the pressure in the flow path 65 upstream of the butterfly valve 72 excessively increases when the value and the time are satisfied.

C. Other Embodiments

(61) (C1) In the above embodiment, the ejection adjustment section 70 includes the butterfly valve 72. In contrast, the ejection adjustment section 70 may include, for example, a shutter mechanism that adjusts the opening area of the flow path 65 by moving across the flow path 65, or a pin-shaped opening and closing mechanism that is provided in the flow path 65 along the direction in which the flow path 65 extends and that adjusts the opening area of the flow path 65 in the vicinity of the nozzle opening 62 by moving toward the nozzle opening 62.

(62) (C2) In the above embodiment, if there is a certain response delay to the change in the angle of the butterfly valve 72 or the change in the rotation speed of the flat screw 40, a command may be transmitted from the control section 300 to the first drive section 74 or the drive motor 32 at a timing in consideration of the response delay. That is, if the timing at which the angle of the butterfly valve 72 is changed or the timing at which the rotation speed of the flat screw 40 is changed actually corresponds to the timing shown in FIG. 8, the command for changing the angle of the butterfly valve 72 may be transmitted prior to the command for changing the rotation speed of the flat screw 40.

(63) (C3) In the above embodiment, as shown in FIG. 8, the control section 300 starts the operation for changing the rotation speed of the flat screw 40 earlier than the operation for changing the opening area of the flow path 65. In contrast, the control section 300 may start the operation of changing the rotation speed of the flat screw 40 and the operation of changing the opening area of the flow path 65 at the same timing.

(64) (C4) In the above described embodiment, the processes of step S18 and step S20 and the processes of step S22 to step S28 in FIG. 6 or FIG. 9 are optional, and the control section 300 may not execute at least a part of these processes.

(65) (C5) In the above embodiment, the control section 300 can adjust the opening area of the flow path 65 at the adjustment speed in two stages, but the adjustment speed may be fixed, or may be adjustable in three stages or steplessly.

(66) (C6) In the above embodiment, the plasticizing section 30 plasticizes the material by the flat screw 40. In contrast, the plasticizing section 30 may plasticize the material by rotating an in-line screw, for example.

(67) (C7) In the above embodiment, the control section 300 controls the rotation angle of the butterfly valve 72 based on the detection value of the angle detection sensor 73. The control section 300 may control the rotation speed of the flat screw 40 based on the detection value of the angle detection sensor 73. By using the angle detection sensor 73, the opening area of the flow path 65 can be accurately detected based on the detection value of the rotation angle of the butterfly valve 72. Therefore, the rotation speed of the flat screw 40 can be accurately controlled.

D. Other Forms

(68) The present disclosure is not limited to the above described embodiments, and can be realized in various configurations without departing from the spirit thereof. For example, the technical features of the embodiments corresponding to the technical features in each aspect described below can be appropriately replaced or combined in order to solve a part or all of the problems described above or in order to achieve a part or all of the effects described above. In addition, unless the technical features are described as essential in the present specification, the technical features can be appropriately deleted.

(69) (1) According to the first aspect of the present disclosure, the three dimensional molding device is provided.

(70) This three dimensional molding device includes a plasticizing section that has a screw in which a groove is formed and that is configured to generate a plasticized material by plasticizing a material by rotating the screw; a nozzle that has a nozzle opening and that is configured to eject the plasticized material; a table on which the plasticized material ejected from the nozzle is deposited; a position changing section configured to change a relative position between the nozzle and the table; an ejection adjustment section that is provided in a flow path, which communicates with the nozzle opening and through which the plasticized material flows, and that is configured to adjust an ejecting amount of the plasticized material from the nozzle by adjusting an opening area of the flow path; and a control section configured to mold a three dimensional molded object in a molding area of the table by controlling the plasticizing section, the position changing section, and the ejection adjustment section, wherein the control section controls a rotation speed of the screw at a first rotation speed when the opening area of the flow path is a first opening area and controls the rotation speed of the screw at a second rotation speed that is faster than the first rotation speed when the opening area of the flow path is a second opening area that is larger than the first opening area.

(71) According to such an aspect, when the ejecting amount of the plasticized material from the nozzle is large, the rotation speed of the screw can be increased to increase the generation amount of the plasticized material, and when the ejecting amount of the plasticized material from the nozzle is small, the rotation speed of the screw can be decreased to decrease the generation amount of the plasticized material. Therefore, it is possible to suppress ejection of an unexpected ejecting amount of plasticized material from the nozzle due to an increase in the pressure in the flow path on the upstream of the ejection adjustment section. Therefore, the molded object can be molded with high accuracy.

(72) (2) The above aspect may be such that the ejection adjustment section includes a drive shaft disposed along a direction intersecting a direction in which the flow path extends, a valve formed in a portion of the drive shaft, a drive section that rotates the valve by rotating the drive shaft, and an angle detection sensor that detects a rotation angle of the valve, the control section is configured to adjust the opening area by controlling the rotation angle of the valve, and the control section is configured to control the rotation speed based on a detection value of the angle detection sensor.

(73) With such a configuration, the rotation speed of the screw can be accurately controlled.

(74) (3) The above aspect may be such that a pressure sensor configured to detect pressure in the flow path upstream of the ejection adjustment section, wherein the control section is configured to decrease the rotation speed of the screw when a pressure exceeding a predetermined value is detected.

(75) According to such an aspect, it is possible to more reliably suppress ejection of an unintended ejecting amount of plasticized material from the nozzle due to an increase in the pressure upstream of the ejection adjustment section.

(76) (4) The above aspect may be such that the control section is configured to decrease the rotation speed of the screw when the opening area remains at or below a predetermined value for a predetermined time or more.

(77) According to such an aspect, it is possible to more reliably suppress ejection of an unintended

ejecting amount of plasticized material from the nozzle due to an increase in the pressure upstream of the ejection adjustment section.

(78) (5) The above aspect may be such that the control section is configured to start an operation for changing the rotation speed of the screw earlier than an operation for changing the opening area of the flow path.

(79) According to such an aspect, it is possible to suppress the occurrence of insufficient supply of the plasticized material when the opening area of the flow path is changed.

(80) (6) The above aspect may be such that the control section is configured to, in adjustment of the opening area, execute either a process of changing the opening area at a first adjustment speed or a process of changing the opening area at a second adjustment speed higher than the first adjustment speed.

(81) According to such a configuration, the opening area of the flow path can be adjusted at the adjustment speed of at least two stages.

(82) (7) The above aspect may be such that the control section is configured to, in adjustment of the opening area, execute either a first process of changing from the second opening area to the first opening area at the first adjustment speed or a second process of changing from the first opening area to the second opening area at the second adjustment speed, and is configured to execute the first process in a case where the relative movement speed of the nozzle is to be decreased and to execute the second process in a case where the relative movement speed of the nozzle is to be increased.

(83) According to such an aspect, when the movement speed of the nozzle is to be increased, the ejecting amount of the plasticized material can be rapidly increased.

(84) (8) According to the second aspect of the present disclosure, there is provided the method for manufacturing the three dimensional molded object. This manufacturing method includes a first step of plasticizing a material by rotating a screw, in which a groove is formed, to generate a plasticized material and a second step of molding a three dimensional molded object by discharging the plasticized material from a nozzle having a nozzle opening, wherein the second step includes a step of adjusting an ejecting amount of the plasticized material from the nozzle by adjusting an opening area of a flow path which communicates with the nozzle opening and through which the plasticized material flows and the first step includes a step of setting the rotation speed of the screw to a first rotation speed when the opening area of the flow path is a first opening area and setting the rotation speed of the screw to a second rotation speed that is higher than the first rotation speed when the opening area of the flow path is second opening area that is larger than the first opening area.

(85) The present disclosure is not limited to the three dimensional molding device and the three dimensional molded object manufacturing method described above, and can be realized by various aspects such as a three dimensional molding system, a computer program, and a non-transitory tangible recording medium in which a computer program is recorded in a computer-readable manner.

Claims

1. A three dimensional molding device comprising: a plasticizing section that has a screw in which a groove is formed and that is configured to generate a plasticized material by plasticizing a material by rotating the screw; a nozzle that has a nozzle opening and that is configured to eject the plasticized material; a table on which the plasticized material ejected from the nozzle is deposited; a position changing section configured to change a relative position between the nozzle and the table; an ejection adjustment section that is provided in a flow path, which communicates with the nozzle opening and through which the plasticized material flows, and that is configured to adjust an ejecting amount of the plasticized material from the nozzle by adjusting an opening area of the flow

path, wherein the ejection adjustment section includes a drive shaft disposed along a direction intersecting a direction in which the flow path extends, a valve formed in a portion of the drive shaft, a drive section that rotates the valve by rotating the drive shaft, and an angle detection sensor that detects a rotation angle of the valve; and a control section configured to mold a three dimensional molded object in a molding area of the table by controlling the plasticizing section, the position changing section, and the ejection adjustment section, wherein the control section: controls a rotation speed of the screw at a first rotation speed when the opening area of the flow path is a first opening area, controls the rotation speed of the screw at a second rotation speed that is faster than the first rotation speed when the opening area of the flow path is a second opening area that is larger than the first opening area, is configured to adjust the opening area by controlling the rotation angle of the valve, and is configured to control the rotation speed based on a detection value of the angle detection sensor.

2. The three dimensional molding device according to claim 1, further comprising: a pressure sensor configured to detect pressure in the flow path upstream of the ejection adjustment section, wherein the control section is configured to decrease the rotation speed of the screw when a pressure exceeding a predetermined value is detected.

3. The three dimensional molding device according to claim 1, wherein the control section is configured to decrease the rotation speed of the screw when the opening area remains at or below a predetermined value for a predetermined time or more.

4. The three dimensional molding device according to claim 1, wherein the control section is configured to start an operation for changing the rotation speed of the screw earlier than an operation for changing the opening area of the flow path.

5. The three dimensional molding device according to claim 1, wherein the control section is configured to, in adjustment of the opening area, execute either a process of changing the opening area at a first adjustment speed or a process of changing the opening area at a second adjustment speed higher than the first adjustment speed.

6. The three dimensional molding device according to claim 5, wherein the control section is configured to, in adjustment of the opening area, execute either a first process of changing from the second opening area to the first opening area at the first adjustment speed or a second process of changing from the first opening area to the second opening area at the second adjustment speed, and is configured to execute the first process in a case where the relative movement speed of the nozzle is to be decreased and to execute the second process in a case where the relative movement speed of the nozzle is to be increased.
